Title of your project

Korea University COSE461 Final Project

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Abstract

An abstract should concisely (less than 300 words) motivate the problem, describe your aims, describe your contribution, and highlight your main finding(s).

This project report is limited to 8 content pages (excluding reference); i.e. you can submit reports with 1-8 pages. Please make your report brief and concise, so that there is no redundant contents. There is no bonus point on submitting full 8 pages report.

1 Introduction

The introduction explains the problem, why it's difficult, interesting, or important, how and why current methods succeed/fail at the problem, and explains the key ideas of your approach and results. Though an introduction covers similar material as an abstract, the introduction gives more space for motivation, detail, references to existing work, and to capture the reader's interest.

2 Related Work

This section helps the reader understand the research context of your work, by providing an overview of existing work in the area.

- You might discuss: papers that inspired your approach, papers that you use as baselines, papers proposing alternative approaches to the problem, papers applying your methods to different tasks, etc.
- This section shouldn't go into deep detail in any one paper (for example, there probably shouldn't be any equations) instead it should explain how the papers relate to each other, and how they relate to your work. If you need to go deep into detail in previous researches with equations, discuss it in the next section.

[1]

3 Approach

This section details your approach(es) to the problem. For example, this is where you describe the architecture of your neural network(s), and any other key methods or algorithms.

- You should be specific when describing your main approaches you probably want to include equations and figures.
- You should also describe your baseline(s). Depending on space constraints, and how standard your baseline is, you might do this in detail, or simply refer the reader to some other paper for the details.
- If any part of your approach is original, make it clear (so I can give you credit!). For models and techniques that aren't yours, provide references.
- If youre using any code that you didn't write yourself, make it clear and provide a reference or link. When describing something you coded yourself, make it clear (so I can give you credit!).
- As youre setting up equations, notation, and the like, be sure to agree on a fixed technical vocabulary (that you've defined, or is well-defined in the literature) before writing and use it consistently throughout the report!

3.1 preprocessing

3.2 Model

We adopt the principle that the encoder of a strong retriever should remain frozen and that any improvement must come from a lightweight, parameter-efficient intervention attached to it. The unmodified frozen ColBERT v2 [2] thus serves as our primary baseline; every Δ NDCG@10 number reported in §4.4 is the paired difference between the corresponding intervention and this baseline on the same query set. This subsection defines the design space of such interventions, organizes the ten variants we examine into three families, and then defines the loss function that drives the final model. Every method discussed below is trained with a single fixed configuration, and each design choice addresses a specific weakness uncovered by the previous one.

Frozen retriever and intervention surface. We use the publicly released ColBERT v2 checkpoint [2] as a frozen encoder. ColBERT represents a query q and a document d as sequences of 128-dimensional token embeddings produced by a BERT-base backbone and a learned linear projection; every token output is L_2-normalized,

$$\hat{h}_t \;=\; \frac{h_t}{\|h_t\|_2} \;\in\; \mathbb{S}^{127} \subset \mathbb{R}^{128}, \tag{1}$$

so the retrieval score reduces to the MaxSim aggregation of token-level cosine similarities,

$$s(q,d) \;=\; \sum_{t\in q} \max_{t'\in d} \cos\bigl(\hat{h}^q_t,\, \hat{h}^d_{t'}\bigr) \;=\; \sum_{t\in q} \max_{t'\in d} \bigl\langle \hat{h}^q_t,\, \hat{h}^d_{t'}\bigr\rangle. \tag{2}$$

Throughout this paper we never modify any weight of the frozen encoder or projection head; we only attach a small trainable module whose forward pass perturbs the contextualized token representation h_t or augments the training loss.

Query slicing on the frozen retriever. For every query x the frozen retriever returns a top-1 document, which lets us partition the query set into two slices,

$$\mathcal{Q}_{\text{hard}} \;=\; \{x : \text{top-1}_{\text{frozen}}(x) \notin \text{qrels}^+(x)\}, \tag{3}$$

$$\mathcal{Q}_{\text{easy}} \;=\; \{x : \text{top-1}_{\text{frozen}}(x) \in \text{qrels}^+(x)\}. \tag{4}$$

The partition is determined entirely by the frozen model and is therefore deterministic and seed-invariant. On SciFact it yields 368 hard and 441 easy training queries (45.5 % / 54.5 %); the test split partitions identically. The two slices are the conceptual axis of our model design: the intervention must lift the hard slice without disturbing the easy slice.

Design space. We organize the interventions into three families. *Family 1* applies a learned direction-subtraction at a single layer; *Family 2* injects low-rank adapters into the query and value projections of every transformer layer; *Family 3* augments the training loss with a representation-anchoring regularizer. We instantiate the three families as ten variants labelled (A)–(J) and report all of them in §4.4.

Family 1: direction-subtraction at a single layer

The simplest intervention removes a direction $v \in \mathbb{R}^{768}$ from the contextualized token representation $h_t^{(\ell)}$ at a target encoder layer ℓ, optionally modulated by a gate $g(\cdot)$:

$$\tilde{h}_t^{(\ell)} \;=\; h_t^{(\ell)} - g(h_t^{(\ell)}) \cdot v. \tag{5}$$

We fix the layer at $\ell = 12$ (the BERT pooler input) throughout and consider four instantiations of the operator on the right-hand side, all trained with the pairwise margin loss of Eq. (12) as the only training signal. Variant (A) computes v once as the difference between the average hard-query and average easy-query token embeddings of the frozen model and applies $\tilde{h} = h - \alpha\,\hat{v}$ with no gate and a fixed scalar α. Variant (B) makes the same vector $v \in \mathbb{R}^{768}$ trainable (768 parameters) and pairs it with either a learned scalar gate $g(h) = \sigma(b)$ or a learned per-token gate $g(h) = \sigma(Wh + b)$. Variant (C) replaces the single direction with a bank of K learned directions $\{v_k\}_{k=1}^{K}$ chosen by a softmax router,

$$\tilde{h}_t = h_t - \sum_{k=1}^{K} \pi_k(h_t)\, v_k, \qquad \pi(h_t) = \mathrm{softmax}(Wh_t + b),$$

sweeping $K \in \{2, 4, 8\}$ as a continuum of mixture sizes. Variant (D) samples v once from $\mathcal{N}(0, I)$ and freezes it, isolating the contribution of magnitude alone (as opposed to a learned direction) as a falsification control.

Family 2: low-rank adapters and triplet-data variants

The methods of Family 1 share two properties that limit them: each operates at a single intervention point, and each compresses the entire correction signal into one (or K) shared 768-dimensional direction(s). To relax both limitations simultaneously we attach Low-Rank Adaptation [3] adapters to the query and value projection matrices of every transformer layer:

$$W \;\mapsto\; W + \tfrac{\alpha}{r}\, BA, \qquad A \in \mathbb{R}^{r \times 768},\; B \in \mathbb{R}^{768 \times r}, \tag{6}$$

with rank $r{=}8$, scaling $\alpha{=}r$, $A_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.02$, and B initialized to zero so that $BA = 0$ and the adapted network is exactly the frozen baseline at step 0. We refer to each $(\text{layer}, \text{projection})$ pair at which an adapter is attached as an *intervention point*; this gives 24 intervention points (12 layers $\times$ $\{q, v\}$) and 294,912 trainable parameters (0.27 % of the 110M-parameter encoder). Family 2 explores three variants that share this adapter architecture and differ only in how the triplet stream that drives the margin loss is constructed.

Variant (E), plain LoRA, optimizes the adapter of Eq. (6) end-to-end with the standard mined triplets and no further regularizer. As we report in §4.4, this setup delivers a large lift on the hard slice but introduces an almost equally large loss on the easy slice, so its aggregate metric appears flat – a phenomenon we will refer to as *performance redistribution* (a measurable, signed transfer of NDCG@10 between the two slices that leaves the aggregate close to zero). Variants (F) and (G) modify the triplet stream as diagnostic probes of the source of this redistribution, keeping the architecture identical to (E). Variant (F) re-scores every mined hard negative with an external E5-Mistral cross-encoder [4] and discards triplets for which the teacher disagrees with the original mining decision, retaining only (q, d^+, d^-) with $\langle e_q^{\text{E5}}, e_{d^+}^{\text{E5}} \rangle > \langle e_q^{\text{E5}}, e_{d^-}^{\text{E5}} \rangle$. Variant (G) replaces the mined hard negative with a uniformly chosen positive document from another query in the same batch,

$$\tilde{d}_i^- \;\sim\; \mathrm{Uniform}\big(\{d_j^+ : j \in [B],\, j \neq i\}\big),$$

where the probability that $\tilde{d}_i^-$ is a true relevant for query i is on the order of one positive over five thousand corpus documents on SciFact, so this is effectively a clean-negative source that strips out the difficulty of the mined hard contrast. The two variants therefore probe different hypotheses about the source of the redistribution: (F) tests whether label noise in the mined HN pool is the cause, while (G) tests whether the difficulty of the mined contrast itself is the cause. Their empirical separation is reported in §4.4; the same separation motivates the loss-side intervention defined next.

Family 3: representation-anchoring regularizer

The trade-off exhibited by plain LoRA can be attacked from the loss side rather than the architecture side. We augment the training objective with a regularizer that prevents easy-query representations from drifting away from the frozen reference. Before stating the regularizer we extend the notation of Eq. (1) to encoder-conditioned token sequences.

For an input token sequence z of length $|z|$, the frozen and the LoRA-adapted encoders both emit 128-dimensional token embeddings $h_t^{\text{frozen}}(z),\, h_t^{\text{LoRA}}(z) \in \mathbb{R}^{128}$ at position $t \in \{1, \ldots, |z|\}$; we apply Eq. (1) component-wise to obtain the L_2-normalized tokens $\hat{h}_t^{\text{frozen}}(z), \hat{h}_t^{\text{LoRA}}(z) \in \mathbb{S}^{127}$. The full L_2-normalized token matrix and its row-wise inner-product (token-token cosine) matrix are

$$H(z) \;=\; \big[\hat{h}_1(z),\, \ldots,\, \hat{h}_{|z|}(z)\big]^\top \in \mathbb{R}^{|z|\times 128}, \qquad \mathrm{Sim}(H) \;=\; H\,H^\top \in \mathbb{R}^{|z|\times|z|}, \tag{7}$$

so that $[\mathrm{Sim}(H(z))]_{t,t'} \;=\; \cos(\hat{h}_t(z),\, \hat{h}_{t'}(z))$. Each easy query $x \in \mathcal{Q}_{\text{easy}}$ is associated with a training triplet $(q_x,\, d_x^+,\, d_x^-)$, where q_x is the query itself, d_x^+ is a paired positive document, and d_x^- is a mined hard negative.

The per-token cosine deviation between the LoRA-adapted and frozen encoders on each role of the triplet is then defined component-wise as

$$\mathcal{R}_{\text{abs}}^{q}(\theta) \;=\; \mathbb{E}_{x\in\mathcal{Q}_{\text{easy}}}\left[\frac{1}{|q_x|}\sum_{t=1}^{|q_x|}\big(1-\cos(\hat{h}_t^{\text{LoRA}}(q_x),\, \hat{h}_t^{\text{frozen}}(q_x))\big)\right], \tag{8}$$

$$\mathcal{R}_{\text{abs}}^{d^+}(\theta) \;=\; \mathbb{E}_{x\in\mathcal{Q}_{\text{easy}}}\left[\frac{1}{|d_x^+|}\sum_{t=1}^{|d_x^+|}\big(1-\cos(\hat{h}_t^{\text{LoRA}}(d_x^+),\, \hat{h}_t^{\text{frozen}}(d_x^+))\big)\right], \tag{9}$$

$$\mathcal{R}_{\text{abs}}^{d^-}(\theta) \;=\; \mathbb{E}_{x\in\mathcal{Q}_{\text{easy}}}\left[\frac{1}{|d_x^-|}\sum_{t=1}^{|d_x^-|}\big(1-\cos(\hat{h}_t^{\text{LoRA}}(d_x^-),\, \hat{h}_t^{\text{frozen}}(d_x^-))\big)\right]. \tag{10}$$

Each $\mathcal{R}_{\text{abs}}^{z}(\theta) \in [0, 2]$ vanishes precisely when the LoRA encoder reproduces the frozen encoder token-by-token on the corresponding role across the easy slice, and grows monotonically with cosine deviation. Alongside these we consider a rotation-invariant variant that anchors only the pairwise token-similarity structure (Eq. (7)) rather than the absolute embedding direction:

$$\mathcal{R}_{\text{rel}}(\theta) \;=\; \mathbb{E}_{x\in\mathcal{Q}_{\text{easy}}}\Big[\big\|\mathrm{Sim}(H_{\text{LoRA}}(q_x)) - \mathrm{Sim}(H_{\text{frozen}}(q_x))\big\|_F^2\Big]. \tag{11}$$

The four regularizers (8)–(11) yield three concrete instantiations of progressively stronger anchoring. Variant (H) uses the rotation-invariant regularizer $\mathcal{R}_{\text{rel}}$ of Eq. (11): the LoRA encoder may freely rotate or reflect the embedding space as long as the pairwise token inner products are preserved. Variant (I) replaces the rotation-invariant regularizer with the stricter rotation-sensitive form $\mathcal{R}_{\text{abs}}^{q} + \mathcal{R}_{\text{abs}}^{d^+}$, fixing the absolute direction of each query token and each positive-document token to its frozen value. Variant (J) is the symmetric completion of (I): since the MaxSim score (2) is symmetric in h^q and h^d, anchoring only the query side and the positive-document side leaves the mined hard negative free to drift through the shared LoRA weights. Adding $\mathcal{R}_{\text{abs}}^{d^-}$ to obtain the regularizer $\mathcal{R}_{\text{abs}}^{q} + \mathcal{R}_{\text{abs}}^{d^+} + \mathcal{R}_{\text{abs}}^{d^-}$ closes this asymmetry and provides a direct empirical probe of whether (I) sits at the ceiling of the anchor family or merely at an artefact of an asymmetric target set.

Final training objective. The combined loss decomposes by query slice. Hard queries supply a pairwise margin loss with mined hard negatives,

$$\mathcal{L}_{\text{margin}}(q, d^+, d^-) \;=\; \max\!\big(0,\; m - s(q, d^+) + s(q, d^-)\big), \quad m{=}0.2, \tag{12}$$

while easy queries supply the anchor terms of Eqs. (8)–(11). The most general form used in this paper is

$$\mathcal{L}(\theta) \;=\; \mathcal{L}_{\text{margin}}\big(\mathcal{Q}_{\text{hard}}; \theta\big) \;+\; \lambda_{\text{dir}}\big(\mathcal{R}_{\text{abs}}^{q} + \mathcal{R}_{\text{abs}}^{d^+}\big) \;+\; \lambda_{\text{neg}}\,\mathcal{R}_{\text{abs}}^{d^-}, \tag{13}$$

with $\lambda_{\text{dir}} = \lambda_{\text{neg}} = 1$ as a single value fixed prior to training. Variant (I) corresponds to $\lambda_{\text{neg}} = 0$, and variant (H) replaces the λ_{dir} term by $\lambda\,\mathcal{R}_{\text{rel}}$.

The two terms in Eq. (13) act on disjoint query subsets, so the gradient field decomposes as

$$\nabla_\theta \mathcal{L} \;=\; \underbrace{\nabla_\theta \mathcal{L}_{\text{margin}}\big|_{\mathcal{Q}_{\text{hard}}}}_{g_{\text{push}}(\theta)} \;+\; \underbrace{\big[\lambda_{\text{dir}}\big(\nabla_\theta \mathcal{R}^{q}_{\text{abs}} + \nabla_\theta \mathcal{R}^{d^+}_{\text{abs}}\big) + \lambda_{\text{neg}}\, \nabla_\theta \mathcal{R}^{d^-}_{\text{abs}}\big]\big|_{\mathcal{Q}_{\text{easy}}}}_{g_{\text{pull}}(\theta)}. \tag{14}$$

The first term pushes the model away from the frozen baseline on hard queries; the second pulls it back on easy queries. Because the anchor is a soft regularizer rather than a hard constraint, the trained model converges to an interior equilibrium point $\theta^\star$ at which the two gradient contributions cancel, $g_{\text{push}}(\theta^\star) + g_{\text{pull}}(\theta^\star) = 0$. We directly measure the corresponding equilibrium expected cosine

$$\mathbb{E}_{x\in\mathcal{Q}_{\text{easy}},\, t}\big[\cos(\hat{h}^{\text{LoRA}}_t(x),\, \hat{h}^{\text{frozen}}_t(x))\big] \;=\; 0.824 \pm 0.005 \tag{15}$$

on SciFact, which lies strictly between the hard-clamp value of 1 (no LoRA drift) and the unanchored LoRA value (large drift): the LoRA representation is permitted to deviate, but only as far as the equilibrium of the two gradient fields allows.

Original contributions and provenance. Of the ten variants defined above, the foundational components are drawn from prior work: ColBERT v2 and its MaxSim score (2) are due to [2]; LoRA (6) is due to [3]; the E5-Mistral cross-encoder used as the teacher of variant (F) is due to [4]; the in-batch easy-negative substitution of variant (G) is a standard technique in contrastive retrieval training, popularised in dense passage retrieval [5]. Variants (A)–(D) instantiate well-known direction-subtraction templates and serve here as baselines and falsification controls. Variant (E) is the standard application of LoRA to ColBERT. The remaining ingredients, to our knowledge, are original to this work: the hard/easy query partition defined by the frozen retriever's top-1 behavior, the per-token cosine anchor regularizers of Eqs. (8)–(10) and their symmetric extension to mined hard negatives (variant J), the joint training objective of Eq. (13) that combines a push loss on the hard slice with a pull regularizer on the easy slice, and the gradient-split analysis of Eqs. (14)–(15) together with the direct measurement of the interior equilibrium $\mathbb{E}[\cos(\hat{h}^{\text{LoRA}}_t, \hat{h}^{\text{frozen}}_t)] = 0.824{\pm}0.005$ across three seeds.

4 Experiments

This section contains the following.

4.1 Data

Describe the dataset(s) you are using (provide references). If it's not already clear, make sure the associated task is clearly described. Being precise about the exact form of the input and output can be very useful for readers attempting to understand your work, especially if you've defined your own task.

4.2 Evaluation method

Describe the evaluation metric(s) you use, plus any other details necessary to understand your evaluation. Some projects will have clear metrics from prior work on given datasets, but we realize that other projects will define their own metrics. If you're defining your own metrics, be clear as to what you're hoping to measure with each evaluation method (whether quantitative or qualitative, automatic or human-defined!), and how it's defined.

4.3 Experimental details

Triplet construction. We mine hard negatives once from the frozen retriever's top-100 pool, taking up to 10 HNs per query and subsampling to a per-seed cap of 9,190 triplets. This cap is the same for every method, so all interventions see exactly the same total amount of supervision and only differ in either (i) the trainable module attached to the encoder or (ii) how the triplet stream is filtered or substituted. Each triplet is routed by query type: if the query lies in $\mathcal{Q}_{\text{hard}}$ the triplet flows to the margin loss of Eq. (12); if it lies in $\mathcal{Q}_{\text{easy}}$ the triplet flows to the anchor terms of Eqs. (8)–(11).

Optimization. We use AdamW with learning rate 5×10^{-5} and weight decay 10^{-4}, batch size 32, for up to 3 epochs with patience-2 early stopping on validation NDCG@10. A $10\,\%$ slice of the training queries is held out as the validation set, and the best LoRA state across epochs is restored at convergence. All experiments share the same optimizer settings; LoRA hyperparameters (r=8, α=r) and the anchor weight (λ_{dir}=λ_{neg}=1) are each a single value fixed prior to training. No per-method sweep is conducted.

Triplet-side variants (F, G). The two triplet-side perturbations defined in §3.2 are implemented at the data layer: (F) discards triplets for which the E5-Mistral cross-encoder disagrees with the original mining decision, retaining $\approx 86\%$ of the SciFact triplets, and (G) replaces the hard negative document with a uniformly sampled positive document from another query in the same batch. Variants F and G are compared to plain LoRA at the same triplet cap of 9,190 (after F's filtering) and the same optimization protocol.

Reproducibility. Every reported number is the mean over three random seeds $\{42, 1337, 2024\}$, with $\pm$ standard deviation. All comparisons against the frozen baseline are reported as paired differences with 10,000-iteration paired bootstrap [6] $95\,\%$ confidence intervals; we use "strict net improvement" to denote the case where the lower CI bound exceeds zero across all three seeds. Corpus scoring is exact brute-force MaxSim over the full corpus (no approximate index), so reported numbers reflect the actual scoring behavior of the model. All code, trained LoRA states, and per-query metrics are released with the paper.

Software and code provenance. Our experimental code is implemented in PyTorch [7] on top of HuggingFace Transformers [8]. The frozen retriever forward pass loads the official `colbert-ir/colbertv2.0` checkpoint and reuses its MaxSim implementation [2]; the E5-Mistral teacher used in variant (F) loads the public `intfloat/e5-mistral-7b-instruct` checkpoint [4]. The LoRA adapter injection of Eq. (6), the anchor regularizers of Eqs. (8)–(11), the slice-routed training loop driven by Eq. (13), the triplet-mining and triplet-substitution pipelines used by variants (F) and (G), and the paired-bootstrap evaluation harness are written by us; no third-party LoRA or contrastive-retrieval library is used. The full source tree and all experiment artefacts are released with the paper.

4.4 Results

Table 1 summarizes Δ NDCG@10 against the frozen baseline for every method introduced in §3.2, decomposed into the hard and easy slices and over three random seeds.

Direction-subtraction interventions do not deliver strict net improvement. Methods A–D from Family 1 of §3.2 attempt to fix hard-query confusion by subtracting a single shared 768-dimensional direction (or a small mixture of such directions) from a single layer. A non-learned mean-difference vector (A) recovers $+0.064$ on the hard subset alone, comparable to a learned single-direction (B), but at the price of a measurable easy-query loss whenever it is measured: the scalar-gated form drops Δ all to -0.020 and the gates saturate at unity. The multi-direction router (C) helps the hard subset roughly proportionally to K, reaching $+0.049$ at K=8, but the additional capacity damages the aggregate metric (the K=8 aggregate is -0.038, the worst of all our learned methods). The random-direction control (D) recovers $+0.011$ on the hard subset with no learned direction at all (aggregate $+0.000 \pm 0.018$, $0/3$ strict), refuting the hypothesis that the magnitude of a single subtraction is by itself the operative quantity. Taken together these methods show that a single intervention point with a single (or low-K mixture of) shared direction(s) is insufficient.

Plain LoRA hides a redistribution. LoRA (E) breaks the limitation of Family 1 by spreading the intervention across 24 separate (layer, projection) points. The aggregate result is $+0.001$ – on the surface a no-op – but the slice decomposition reveals that this hides a $+0.104$ lift on the hard slice purchased at the cost of a -0.085 loss on the easy slice. The redistribution is quantitatively exact. Writing the slice fractions on the SciFact test split as

$$w_{\text{hard}} = \frac{|\mathcal{Q}^{\text{test}}_{\text{hard}}|}{|\mathcal{Q}^{\text{test}}|} = \frac{137}{300} \approx 0.457, \qquad w_{\text{easy}} = \frac{|\mathcal{Q}^{\text{test}}_{\text{easy}}|}{|\mathcal{Q}^{\text{test}}|} = \frac{163}{300} \approx 0.543, \tag{16}$$

Table 1: Δ NDCG@10 against the frozen ColBERT v2 baseline on SciFact (test split, 300 queries; hard n=137, easy n=163). Mean $\pm$ std over three seeds where applicable; "N/A" denotes a slice not measured for that variant (Family 1 single-seed sweeps generally report only the slice that the method targets). ✓indicates a strict net improvement (the lower bound of the 95 % paired-bootstrap CI exceeds zero) in k of 3 seeds; "N/A" under *strict* indicates that the variant does not have a measured Δ_{all}. Group separators correspond to the three intervention families of §3.2. **Bold** marks the best Δ all and the corresponding method.

Method	Δ all	Δ hard	Δ easy	strict
Family 1: direction-subtraction at a single layer				
(A) mean-difference, α=10	N/A	$+0.064$	N/A	N/A
(B) single direction + scalar gate	-0.020	N/A	N/A	$0/3$
(B) single direction + per-token gate	N/A	-0.003	N/A	N/A
(C) K=2 mixture	$+0.015$	$+0.039$	N/A	$0/3$
(C) K=4 mixture	$+0.015$	$+0.045$	N/A	$0/3$
(C) K=8 mixture	-0.038	$+0.049$	N/A	$0/3$
(D) random direction, α=10	$+0.000 \pm 0.018$	$+0.011$	N/A	$0/3$
Family 2: low-rank adapters and triplet-data variants				
(E) plain LoRA r=8	$+0.001 \pm 0.003$	$+0.104 \pm 0.012$	-0.085 ± 0.010	$0/3$
(F) + false-negative removal	-0.004 ± 0.005	$+0.080 \pm 0.011$	-0.073 ± 0.010	$0/3$
(G) + in-batch easy negatives	$+0.021 \pm 0.004$ ✓	$+0.065 \pm 0.010$	-0.017 ± 0.006	$3/3$
Family 3: representation-anchoring regularizer				
(H) relational anchor	$+0.029 \pm 0.005$	$+0.101 \pm 0.010$	-0.031 ± 0.018	$2/3$
(I) per-token anchor on $\{q, d^+\}$	$+\mathbf{0.030 \pm 0.002}$ ✓	$+0.092 \pm 0.007$	-0.021 ± 0.003	$3/3$
(J) symmetric anchor on $\{q, d^+, d^-\}$	$+0.028 \pm 0.001$ ✓	$+0.077 \pm 0.003$	-0.014 ± 0.001	$3/3$

and using the linearity of NDCG@10 averaging over disjoint slices, the slice-additive decomposition

$$\Delta_{\text{all}} \;=\; w_{\text{hard}}\,\Delta_{\text{hard}} + w_{\text{easy}}\,\Delta_{\text{easy}} \quad\Longleftrightarrow\quad \Delta_{\text{easy}} \;=\; \frac{\Delta_{\text{all}} - w_{\text{hard}}\,\Delta_{\text{hard}}}{w_{\text{easy}}} \tag{17}$$

predicts $\Delta_{\text{easy}} \approx (0.001-0.457\cdot(+0.104))/0.543 \approx -0.086$, matching the observed -0.085 within one standard deviation. The intervention is therefore not "inert"; it actively moves probability mass between the two slices.

Isolating the cause of the redistribution. The triplet-side variants F and G probe two candidate explanations. False-negative removal (F) leaves a comparable redistribution: Δ hard drops slightly to $+0.080$, Δ easy improves only marginally to -0.073, and the aggregate is still negative (-0.004). Label noise in the mined hard-negative pool is therefore not the operative cause – removing the noisy fraction does not undo the redistribution. In-batch easy negative substitution (G) tells the opposite story: by replacing the mined hard negative with a clean easy negative the easy-query loss drops by an order of magnitude to -0.017, the hard-query lift is roughly halved to $+0.065$, and the aggregate crosses into the first strict net improvement of the paper at $+0.021$ ($3/3$ strict). The pair (F, G) thus isolates the difficulty of the mined hard contrast – not its label noise – as the mechanism producing the redistribution.

Representation anchoring delivers strict net improvement. Adding the easy-side anchor of Eqs. (8)–(13) to plain LoRA – without touching the triplet stream – recovers the full hard-query lift of LoRA while neutralizing most of the easy-query damage. Variant (I), the per-token anchor on $\{q, d^+\}$, achieves Δ all$= +\, 0.030 \pm 0.002$ ($3/3$ strict), preserving 88 % of plain LoRA's hard lift ($+0.092$ versus $+0.104$) and cutting the easy loss by 75 % (-0.021 versus -0.085). The rotation-invariant variant (H) lands within statistical error of (I) at $+0.029\pm0.005$ ($2/3$ strict), so the absolute and relational geometric forms of the anchor produce statistically indistinguishable aggregate results and no specific algebraic form of the regularizer is preferred. The symmetric extension to the negative side (J) then tests whether closing the asymmetry of (I)'s target set yields a further improvement; it does not. Variant (J) shrinks the easy-query loss further to -0.014 and reduces LoRA aggressiveness ($\|B\|_{\text{tot}}$ falls from 1.34 to 1.24), but the hard-query lift drops by an almost exactly compensating amount to $+0.077$, leaving the aggregate at $+0.028 \pm 0.001$ – statistically indistinguishable from (I) at $+0.030$. The three regularizers therefore land at the same $(\Delta\,\mathrm{hard}, \Delta\,\mathrm{easy})$ operating point under our experimental conditions.

Internal-geometry measurements at the equilibrium point. The convergence of three distinct regularizers to a common operating point is also visible in the trained encoder's internal geometry. The equilibrium expected cosine of Eq. (15) is 0.824 ± 0.005 across seeds, almost identical between variants (H) and (I). To quantify how much of the original representation diversity is preserved we measure the *effective rank* [9] of the easy-query token cloud. For a token matrix $\mathbf{H} \in \mathbb{R}^{N \times d}$ with singular values $\sigma_1, \ldots, \sigma_{\min(N,d)}$, the effective rank is the exponentiated Shannon entropy of the normalized squared spectrum,

$$\text{eff-rank}(\mathbf{H}) \;=\; \exp\!\left(-\sum_i p_i \log p_i\right), \qquad p_i \;=\; \frac{\sigma_i^2}{\sum_j \sigma_j^2}, \tag{18}$$

which equals the nominal rank when the spectrum is flat and collapses to 1 when all mass concentrates on a single mode. The token-level effective rank rises from 1.58 under plain LoRA to 9.01 ± 0.36 under variant (I), recovering most of the way toward the frozen value of 55.13. These two measurements – the equilibrium cosine of Eq. (15) and the effective rank of Eq. (18) – link the observed saturation to a single equilibrium point of the two-component gradient field $\nabla_\theta \mathcal{L} = g_{\text{push}} + g_{\text{pull}}$ defined in Eq. (14).

Summary of findings. Going from the simplest direction-subtraction interventions to the anchored LoRA model, the experimental sequence yields three conclusions. (i) A single shared direction at a single intervention point cannot deliver a strict net improvement (variants A–D). (ii) LoRA at all 24 intervention points obtains the hard-query lift but at the cost of an almost equal easy-query loss, and the triplet-side controls (F, G) trace this loss to the difficulty of the mined hard negatives rather than to label noise in the pool. (iii) An easy-side representation-anchor regularizer closes the loss, achieving the first $3/3$ strict net improvement of the paper at $\Delta\,\mathrm{all} = +0.030$, and three mathematically independent regularizers (H, I, J) converge to the same operating point in our experiments, with $+0.030$ acting as a practical upper bound under the fixed configuration used here.

5 Analysis

Your report should include *qualitative evaluation*. That is, try to understand your system (e.g. how it works, when it succeeds and when it fails) by inspecting key characteristics or outputs of your model.

6 Conclusion

Summarize the main findings of your project, and what you have learnt. Highlight your achievements, and note the primary limitations of your work. If you like, you can describe avenues for future work.

After you are done writing this report, save it as a .pdf file, and submit through BlackBoard. You don't have to submit a copy per person; submitting once per team is fine.

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A Appendix: Team contributions

The stuff here does not count towards the 8 page limit. If you are a multi-person team, you can write a brief summary of what each team member did for the project (about 1 or 2 sentences per person). For almost all teams, it will have no effect (i.e. team members all receive same grade), but for teams with considerably unequal contribution, I may investigate and/or give different grades to team members.